JOHN HAROLD D. DOTON

AI/ML Engineer | Computer Science Student

harolddoton@gmail.com | 0936-029-6134 | San Rafael, Bulaon, San Fernando, Pampanga, Philippines

linkedin.com/in/harolddoton • github.com/Haruuowo • haroldportfolio.vercel.app

Technical Skills

- **Languages:** Python, JavaScript/TypeScript, Java, Go, Kotlin
- **AI / ML:** Transformer fine-tuning (Whisper/HuggingFace), transfer learning, RAG, TTS pipelines, scikit-learn
- **Mobile:** Android (Accessibility Service API), Kotlin
- **Web:** HTML, CSS, JavaScript, React, Node.js, REST APIs
- **Tools & Platforms:** Git, GitHub, Google Colab, Node.js, LibreOffice, whisper.cpp, VS Code, Antigravity
- **Concepts:** Data Structures & Algorithms (NeetCode 150), ML theory fundamentals, offline-first architecture

Experience

Internal Officer Angeles City, Pampanga
Google Developer Group (GDG) HAU2024 – Present

- Spearheaded technical workshops and coordinated community hackathons, facilitating hands-on learning for 100+ student developers in cloud and mobile technologies.

Projects

HandsFree (Capstone): Offline Voice-Controlled Android App2024 – Present
Lead Developer & Researcher | Kotlin, Whisper, Android Accessibility Service API

- Designed an offline overlay Android app capturing voice commands in Filipino, English, and Taglish with zero network dependency and zero data transmission.
- Architected a translation normalization layer, a transformer-based ASR fine-tuning pipeline (OpenAI Whisper), and an intent classifier dispatching simulated gestures via the Accessibility Service API.
- Fine-tuned on a Tagalog-English-Taglish corpus (Mozilla Common Voice, Google FLEURS, LibriSpeech) using cross-entropy loss and gradient-based optimization.
- Implemented noise suppression (WebRTC/RNNoise) and confidence thresholding for out-of-vocabulary rejection.
- Delivered five core functions: App Launching, Smart Search, Messaging, Accepting Calls, and Scheduling.

Luna (AI VTuber)2026 – Present
Solo Developer | LLM, TTS, RAG

- Building a real-time AI virtual streamer powered by an LLM/TTS/RAG stack with Twitch and TikTok Live integration.
- Architecting a RAG memory layer and speech synthesis pipeline for interactive, personality-consistent live responses.

SmartSort (Intelligent File Organizer)2026
Solo Developer | Python, PyWebView, HTML/CSS/JS, SQLite

- Built a desktop file organizer with a modern dashboard UI that classifies files by content signatures instead of extensions, with EXIF-aware photo sorting by year and month.
- Implemented a YAML-based custom rule engine, exact-duplicate detection, dry run mode, and real-time folder watching for automatic organization.
- Designed a SQLite history system that logs every file move so any operation can be undone; added unit tests with pytest and packaged a standalone Windows executable with PyInstaller.

PhishPolluter (Anti-Phishing Data Pollution Tool)2026
Solo Developer | Python

- Built a Python tool demonstrating web automation, rate limiting, rotating user-agents, and human-like request pacing as a study in ethical offensive security.
- Added ethical guardrails: configurable submission limits, disclaimer documentation, and a reporting workflow (Google Safe Browsing, DICT-CICC, registrar abuse channels).

ReactionCam (Real-Time Gesture & Expression Detector)2024
Solo Developer | Python, MediaPipe, OpenCV, Pygame

- Developed a real-time computer vision app using MediaPipe and OpenCV to identify 8 distinct facial affects and manual gestures from live video.
- Engineered a synchronized pipeline integrating facial landmark triangulation and hand pose estimation with event-driven audio-visual feedback via Pygame.

Education

Holy Angel University Angeles City, Pampanga
Bachelor of Science in Computer Science, School of ComputingExpected 2025

- Relevant Coursework: Data Structures & Algorithms, Machine Learning, Mobile Development, Software Engineering